Nonlinear Equations
Part 2

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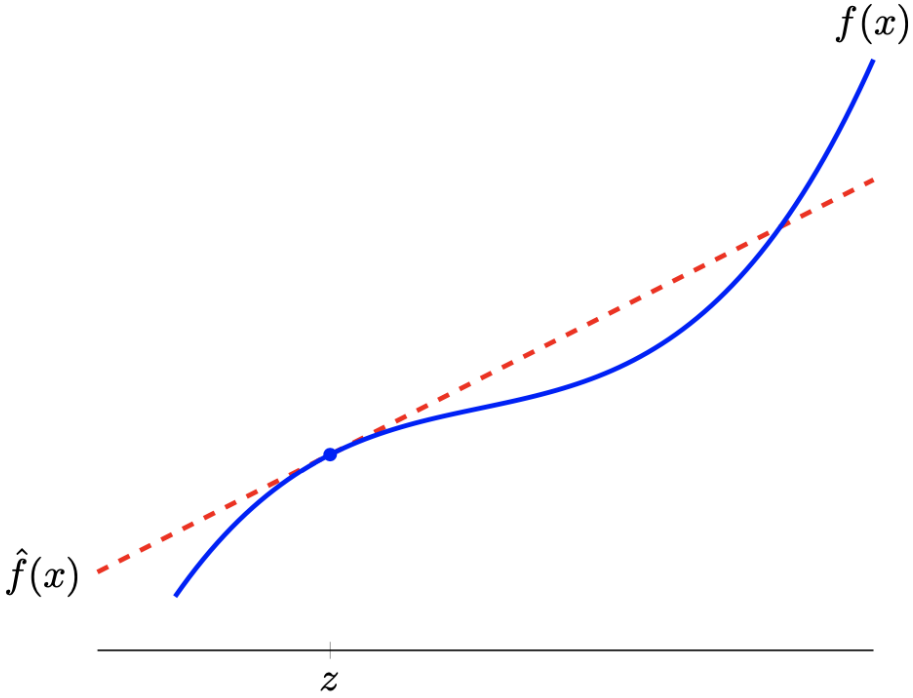
Linear Approximation



Background: linear Taylor approximation

- the first-order (linear) Taylor approximation about z is

$$\hat{f}(x) = f(z) + f'(z)(x - z);$$

- the error in this approximation is

$$|f(x) - \hat{f}(x)| < \tfrac{1}{2} M (x - z)^2,$$

$$M = \max_{z<\xi<x} f''(\xi)$$

- sometimes, $\hat{f}$ is written as

$$\hat{f}(x; z) = f(z) + f'(z)(x - z).$$

Newton's method

- core idea: replace f by a linear Taylor approximation $\hat{f}$ centered at $x^{(k)}$:

$$f(x) \approx \hat{f}(x; x^{(k)}) = f(x^{(k)}) + f'(x^{(k)})(x - x^{(k)});$$

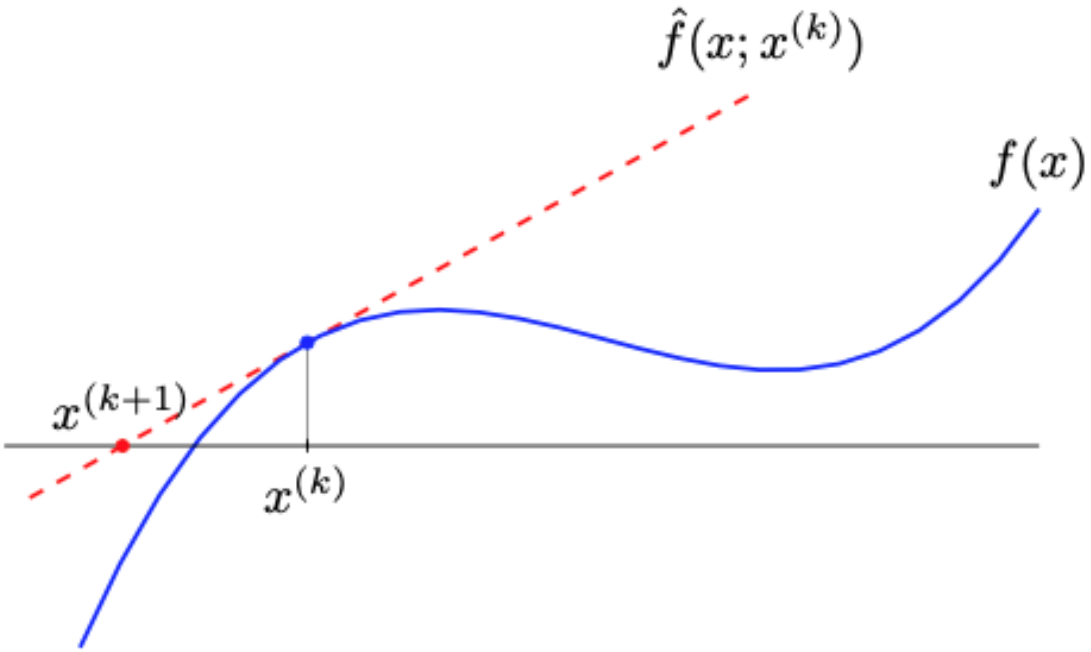
- set $x^{(k+1)}$ as the solution of $\hat{f}(x; x^{(k)}) = 0$:

$$f(x^{(k)}) + f'(x^{(k)})(x^{(k+1)} - x^{(k)}) = 0;$$

- if $f'(x^{(k)}) \neq 0$,

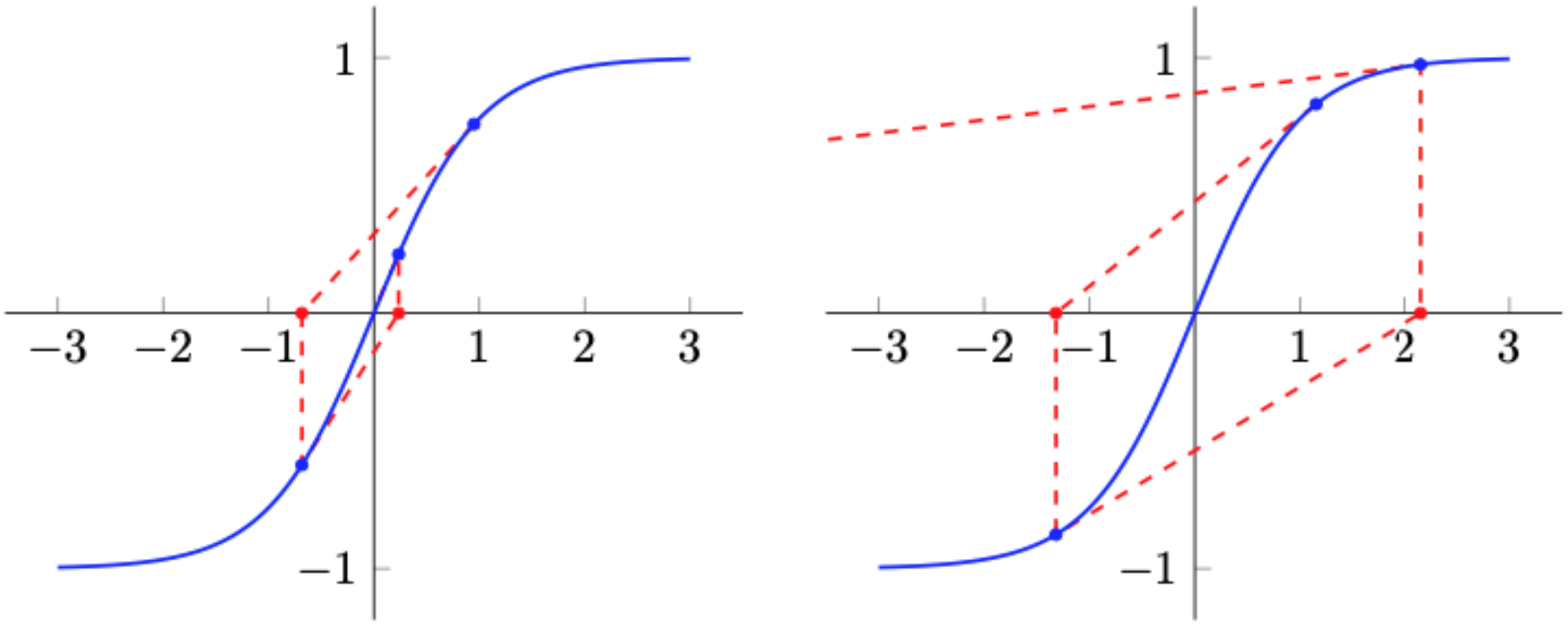
$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}.$$

Newton's Method Illustrated



Does Newton's Method Always Converge?

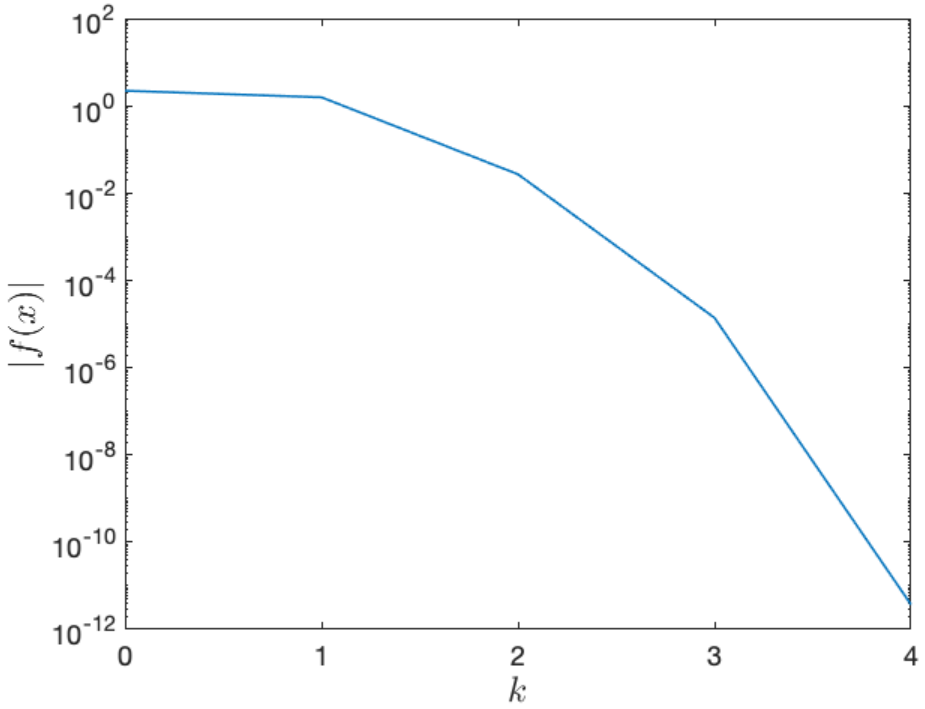
- Newton's method converges when the starting point $x^{(0)}$ is "sufficiently close" to r:



- Newton's method fails if it encounters a point $x^{(k)}$ where $f'(x^{(k)}) = 0$;
- yet, it is the main workhorse for solving nonlinear equations!

Newton's Method: Rate of Convergence

- let f have continuous f'' in $[a, b]$ where there is a simple root r ($f'(r) \neq 0$);
- then there is a Δ such that starting with $x^{(0)}$ anywhere in the interval $[r - \Delta, r + \Delta]$, Newton's method converges *quadratically*:



Matlab Implementation

```
function [sol, his] = newton(f, fprime, x0, tol)
history = [];
x = x0;
while abs(f(x)) > tol
his = [his; x];
x = x - f(x) / fprime(x);
end
sol = x;
his = [his; sol];
end
```

example:

```
f1 = @(x) x - x.^(1/3) - 2;
f1p = @(x) 1 - (1/3) * x.^(-2/3);
[x, hist] = newton(f1, f1p, 0.5, 1e-12);
semilogy(0:length(h)-1, abs(f1(h))
```

More Food for Thought

What if we use Newton's method to find the roots of

$$x^2 = 0?$$

The Secant Method

- Newton's method needs f';
- the Secant Method circumvents the need for explicitly evaluating this derivative, by substituting f' by its finite difference approximation:

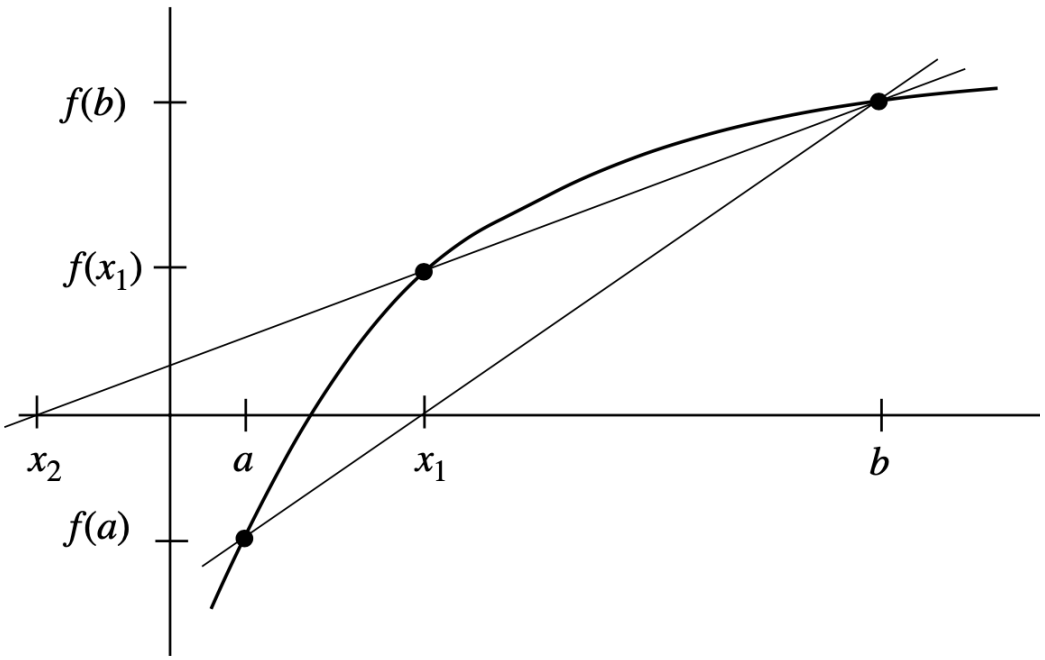
$$f'(x^{(k)}) \approx \frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$$

- the secant iteration or quasi-Newton iteration is

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})\,(x^{(k)} - x^{(k-1)})}{f(x^{(k)}) - f(x^{(k-1)})};$$

- the secant method needs *two* starting iterates.

The Secant Method Illustrated

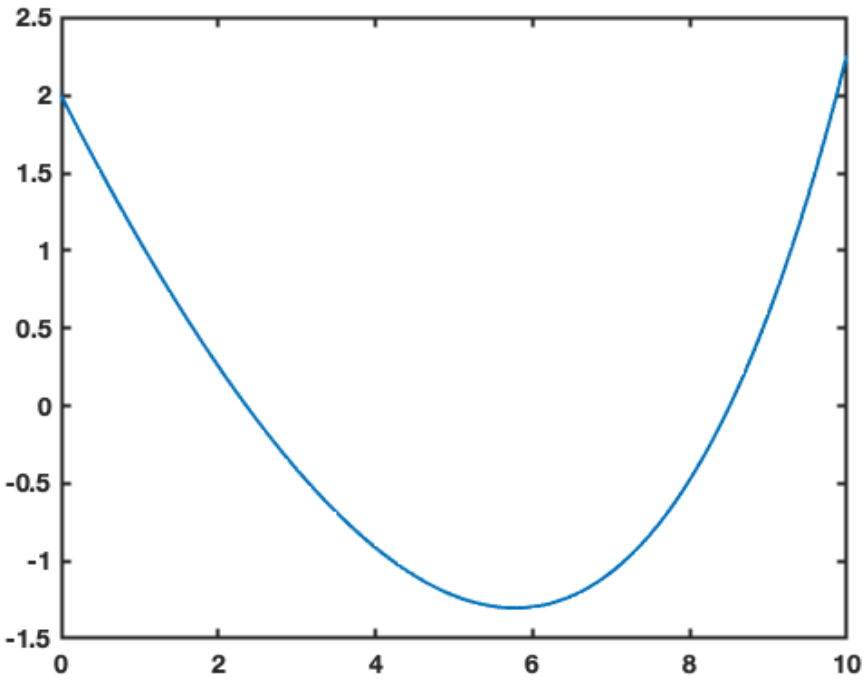


The Secant Method: Rate of Convergence

- as with Newton's method, there is no guarantee that the secant method converges;
- initial iterates must be "close" to the root;
- superlinear convergence with $r \approx 1.6$;
- much faster than the bisection method but not quite as fast as the Newton's iteration;
- variations on the basic idea can make it more robust but will slow down the convergence to linear rate.

Comparison

$$\exp(x/4) + \exp(-x/4) - x = 0$$



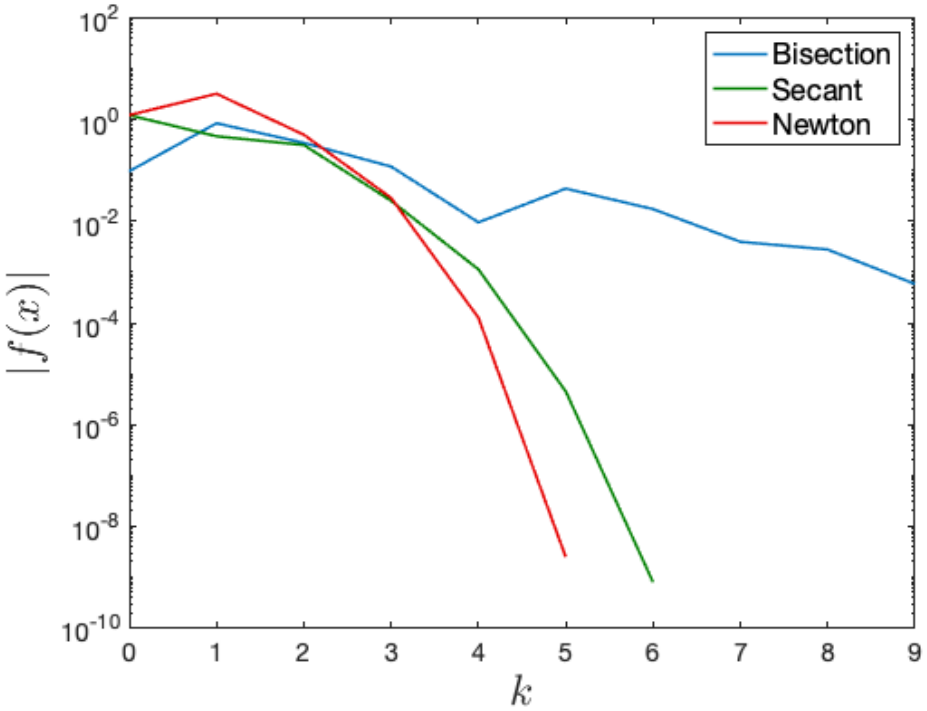
Comparison

```
f2 = @(x) exp(x/4) + exp(-x/4) - x;
f2p = @(x) 0.25*exp(x/4) - 0.25*exp(-x/4) - 1;

[x1, h1] = bisect(f2, 0, 5, 1e-3);
% needs the interval [0, 5]
[x2, h2] = secant(f2, 0, 5, 1e-8);
% needs 2 starting points 0, 5
[x3, h3] = newton(f2, f2p, 5, 1e-8);
% needs a starting point and f'

semilogy(abs(f2(h1))
hold on
semilogy(abs(f2(h2))
semilogy(abs(f2(h3))
```

Comparison


Bisection
Secant
Newton
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
$|f(x)|$
0
1
2
3
4
5
6
7
8
9
k

Fixed Point Iteration

A fixed point of a function g is a point $x^\star$ satisfying

$$x^\star = g\left(x^\star\right).$$

The idea is that $f(x) = 0$ can be written as $x = g(x)$ and that

$$x^{(k+1)} = g(x^{(k)}), \quad k = 0, 1, 2, \ldots$$

with a given $x^{(0)}$ converges to a fixed point. This algorithm is called the Fixed Point Iteration.

Existence and Uniqueness of a Fixed Point

1. if $g \in C[a, b], g(a) \geq a, g(b) \leq b$, then there exists a fixed point $x^\star \in [a, b]$; the proof is based on the Intermediate Value Theorem;
2. in addition, if g' exists and there exists $\rho < 1$ such that

$$|g'(x)| \leq \rho,$$

for all $x \in (a, b)$, then $x^\star$ is unique in $[a, b]$; the proof is by contradiction and based on Taylor truncation error.

Convergence of the Fixed Point Iteration

1. question: under the same assumptions, does the fixed point iteration converge to $x^\star$?
2. $|x^{(k+1)} - x^\star| = |g(x^{(k)}) - g(x^\star)| = |g'(\xi)(x^{(k)} - x^\star)|$, with ξ between $x^{(k)}$ and $x^\star$;
3. then
$$|x^{(k+1)} - x^\star| \leq \rho |x^{(k)} - x^\star|,$$
which represents a contraction by the factor ρ;
4. moreover,
$$|x^{(k+1)} - x^\star| \leq \rho^{k+1} |x^{(0)} - x^\star|;$$
5. as $k \to +\infty$, $\rho^k \to 0$, establishing convergence of the fixed point iteration.

Rate of Convergence of the Fixed Point Iteration

$$|x^{(k+1)} - x^\star| \leq \rho |x^{(k)} - x^\star|$$

1. linear rate of convergence;
2. contrary to the bisection method, where $C = 1/2$ does not depend on f, ρ depends on $|g'|$ and hence on f;
3. how many iterations to reduce the error by a factor of 10, an order of magnitude?

$$|x^{(k)} - x^\star| \approx \rho^k |x^{(0)} - x^\star| \approx 0.1 |x^{(0)} - x^\star|$$

$$k = -\frac{1}{\log_{10} \rho}.$$

4. note that the convergence properties of the iteration depend on the choice of g.

Zeros of Polynomials: an Eigenvalue Problem

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

- the eigenvalues of A and the zeros of its characteristic polynomial $p(\lambda)$ are the same:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -c_0 & -c_1 & -c_2 & -c_3 \end{bmatrix},$$

$$p(\lambda) = \lambda^4 + c_3\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0;$$

- we can find all the roots and we don't need to iterate; but only applicable for polynomials.

Nonlinear Equations in $\mathbb{R}^n$

$$
\begin{aligned}
f_1(x_1, x_2, \cdots, x_n) &= 0 \\
f_2(x_1, x_2, \cdots, x_n) &= 0 \\
&\vdots \\
f_n(x_1, x_2, \cdots, x_n) &= 0
\end{aligned}
$$

- $x = [x_1 \quad x_2 \quad \ldots \quad x_n]$;
- $f_i(x) = 0$ is the ith equation and $f_i(x)$ is the ith residual;
- $f = [f_1 \quad f_2 \quad \ldots \quad f_n]$;
- we can write the set of nonlinear equations in canonical form: $f(x) = 0$;
- when the $f_i(x)$ are linear, $f(x) = 0$ reduces to set of n linear equations.

Example: $n = 2$

$$\sin(x_1) + x_2 + 4 = 0$$
$$x_1^2 - x_1 x_2 = 0$$

```
myf = @(x) [sin(x(1))+x(2)+4; x(1)^2-x(1)*x(2)];
```

goal: to find a 2×1 vector `xs` such that `myf(xs)` returns $\mathbf{0}$ within some tolerance

Nonlinear Equations

- solving nonlinear equations is, in general, much harder than solving linear equations;
- even determining if a solution exists is hard;
- even in the simpler univariate case, stating from different initial guesses or using different initial intervals gives us different solutions (or no solutions);
- for the nonlinear case, there is no analog of LU or QR.

Nonlinear Equations in $\mathbb{R}^n$

- for the univariate case: bisection, Newton's and secant methods;
- the bisection method can be generalized to higher dimensions but extensions are non-trivial and result in slow algorithms as n increases;
- Broyden's method: it is the generalization of the secant method but we will not consider it;
- we will focus on generalizing Newton's method to higher dimension to benefit from its rapid quadratic convergence.

Newton's Method in $\mathbb{R}^n$

- Newton's method relies on successive linearizations of the nonlinear function;
- we will follow a similar procedure for

$$f : \mathbb{R}^n \to \mathbb{R}$$

and

$$f : \mathbb{R}^n \to \mathbb{R}^n.$$

Linear Approximation of $f : \mathbb{R}^n \to \mathbb{R}$

- $f : \mathbb{R}^n \to \mathbb{R}$;
- the gradient of f at z is

$$\nabla f(z) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(z) & \dots & \frac{\partial f}{\partial x_n}(z) \end{bmatrix};$$

- the first-order Taylor approximation of f at z is

$$\hat{f}(x; z) = f(z) + \frac{\partial f}{\partial x_1}(z)\,(x_1 - z_1) + \cdots + \frac{\partial f}{\partial x_n}(z)\,(x_n - z_n);$$

$$\hat{f}(x; z) = f(z) + \nabla f(z)(x - z);$$

- $\hat{f}(x; z)$ is a linear function of x and it is the linear approximation of f at z.

Linear Approximation of $f : \mathbb{R}^n \to \mathbb{R}^n$

- $f : \mathbb{R}^n \to \mathbb{R}^n$;
- each component of f, $f_i(x)$, is a scalar function and can be linearly approximated at z by

$$\hat{f}_i(x; z) = f_i(z) + \nabla f_i(z)(x - z);$$

- then,

$$f(x) \approx \begin{bmatrix} \hat{f}_1(x, z) \\ \vdots \\ \hat{f}_n(x; z) \end{bmatrix} = \begin{bmatrix} f_1(z) \\ \vdots \\ f_n(z) \end{bmatrix} + \begin{bmatrix} \text{—} & \nabla f_1(z) & \text{—} \\ & \vdots & \\ \text{—} & \nabla f_n(z) & \text{—} \end{bmatrix} \begin{bmatrix} x_1 - z_1 \\ \vdots \\ x_n - z_n \end{bmatrix}$$

$$f(x) \approx \hat{f}(x; z) = f(z) + Df(z)(x - z);$$

- where $Df(z)$ is the $n \times n$ *Jacobian* matrix:

$$Df(z)_{ij} = \frac{\partial f_i}{\partial x_j}(z), \quad i, j = 1, \ldots, n.$$

Newton's Method for $f : \mathbb{R}^n \to \mathbb{R}^n$

- at each iteration k, we form the linear approximation $\hat{f}$ of f at current iterate $x^{(k)}$, given by the Taylor approximation:

$$\hat{f}(x; x^{(k)}) = f(x^{(k)}) + Df(x^{(k)})(x - x^{(k)});$$

- the next iterate, $x^{(k+1)}$, is found by solving the linearized equations

$$f(x^{(k)}) + Df(x^{(k)})(x - x^{(k)}) = 0;$$

- then,

$$x^{(k+1)} = x^{(k)} - Df(x^{(k)})^{-1} f(x^{(k)}).$$

Newton's Algorithm

- while (not converged), iterate as follows:
  - compute Jacobian at $x^{(k)}$: $J^{(k)} = Df(x^{(k)})$;
  - solve $J^{(k)}\Delta x = -f(x^{(k)})$
  - update $x^{(k)}$ with Δx : $x^{(k+1)} = x^{(k)} + \Delta x$.

Example 1

$$x_1^2 - 2x_1 - x_2 + 1 = 0$$
$$x_1^2 + x_2^2 - 1 = 0$$

- compute residual/vector function:

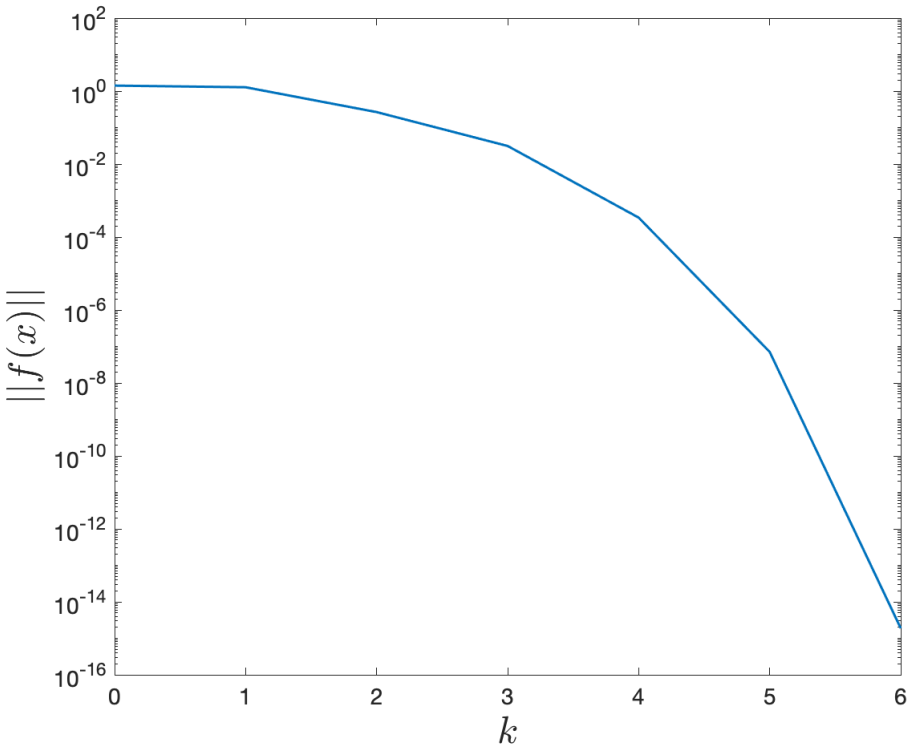
$$f(x) = \begin{bmatrix} x_1^2 - 2x_1 - x_2 + 1 \\ x_1^2 + x_2^2 - 1 \end{bmatrix};$$

- compute Jacobian:

$$Df(x) = \begin{bmatrix} 2x_1 - 2 & -1 \\ 2x_1 & 2x_2 \end{bmatrix};$$

- initial guess $x^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Convergence



Example 2

- nonlinear BVP:

$$u''(x) + e^{u(x)} = 0, \quad 0 < x < 1,$$
$$u(0) = u(1) = 0;$$

- discretize and turn the ODE into a system of nonlinear algebraic equations;
- divide $[0, 1]$ into n equal subintervals of length $h = \frac{1}{n}$;
- $n + 1$ unknown values of u at $x_i = (i - 1)h$: $u_i = u(x_i)$;
- the discretization yields

$$u_1 = 0$$
$$\frac{1}{h^2}(u_{i+1} - 2u_i + u_{i-1}) + e^{u_i} = 0 \quad \text{for all interior } i$$
$$u_{n+1} = 0;$$

Example 2

- Jacobian:

$$Df(u) = \begin{bmatrix} 1 & 1 & & & & \\ 1 & \frac{-2}{h^2} + e^{u_1} & 1 & & & \\ & 1 & \frac{-2}{h^2} + e^{u_2} & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & \frac{-2}{h^2} + e^{u_n} & 1 \\ & & & & 1 & 1 \end{bmatrix}$$

- $u^{(0)} = \mathbf{0}$, for example;
- remove the first and last rows and columns and work with the interior part of the Jacobian (i.e, only the interior $n-1$ values are unknowns).

Newton's Method

- Newton's method converges quadratically to a root r:

$$||r - x^{(k+1)}|| \leq M_1 ||r - x^{(k)}||^2, \quad ||f(x^{(k+1)})|| \leq M_2 ||f(x^{(k)})||^2$$

  for $M_1, M_2 > 0$ constants;
- in problem set, you will compute the observed convergence rate to verify;
- some disadvantages:
  - need to solve a (possibly large) system of linear equations,
    - but we have fast linear solvers;
  - need to compute derivatives,
    - but we can do so automatically;
  - may not converge unless started from a close initial guess,
    - but we can modify the basic method slightly to globalize it and ensure convergence;